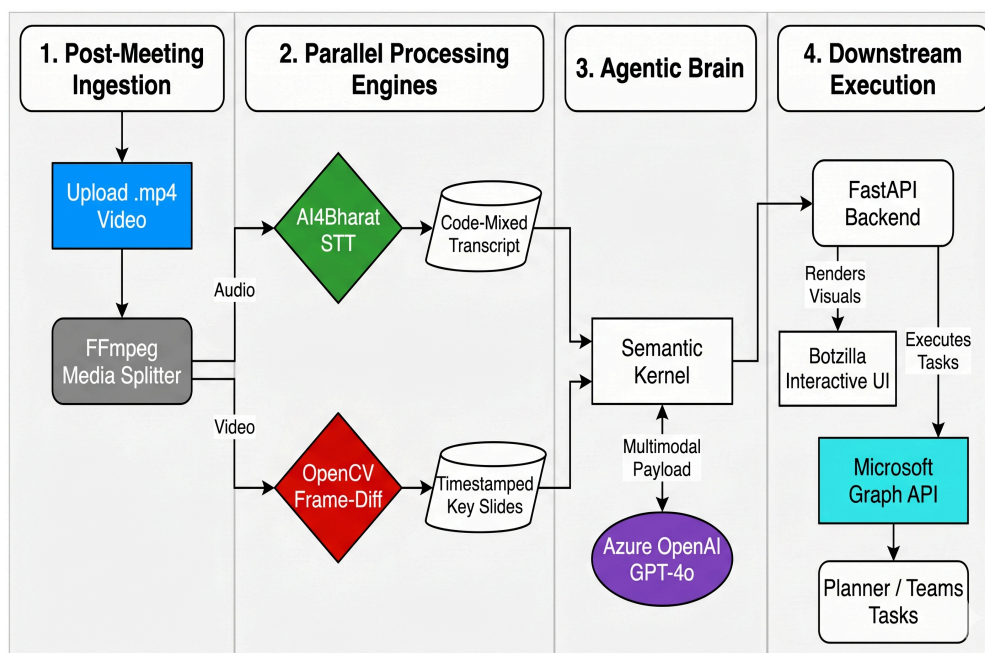


## THE PROBLEM STATEMENT DEMYSTIFIED

In modern corporate environments, millions of hours are wasted re-listening to meeting recordings just to capture decisions and action items. Important tasks lose their owners, deadlines are missed, and critical information gets buried.

Capgemini is asking for an **Autonomous Agentic Workflow**. They do not want a basic text summarizer. They want an intelligent agent that ingests raw meeting data, guarantees 100% faithful extraction (zero hallucinations), respects enterprise data privacy, and actively executes downstream tasks (like setting up calendar events or project boards) through software connectors.

## HOW BOTZILLA WORKS: END-TO-END CORE WORKFLOW



- 1. Ingestion & Media Demuxing:** The user uploads a post-meeting video file ( .mp4 ) into the Botzilla web dashboard. The backend utilizes **FFmpeg** to immediately split the file into two independent tracks: a raw audio file and a raw video stream.
- 2. Visual Slide Extraction:** The video stream is processed asynchronously via **OpenCV**. Instead of scanning every single frame, a frame-differencing algorithm computes pixel deltas. When a major shift occurs (e.g., a

presenter flips a PowerPoint slide), OpenCV captures a high-resolution screenshot and flags the exact millisecond timestamp.

3. **Bilingual Speech-to-Text:** The audio track is fed directly into the **AI4Bharat model** (such as *Chitrlekha* or *IndicConformer* frameworks). This handles regional speech anomalies and code-mixed inputs (Hinglish/Hindi-English switching) seamlessly, outputting a precise, time-coded text transcript.
4. **Agentic Orchestration: Semantic Kernel** serves as the central brain. It packages the AI4Bharat text transcript and maps the OpenCV visual frames to their corresponding timestamps. This unified multimodal payload is sent to **Azure OpenAI (GPT-4o)**.
5. **Downstream Execution:** The agent parses the data into a deterministic JSON layout containing tasks, owners, and due dates. It translates this JSON into automated payload instructions for the **Microsoft Graph API**, instantly preparing Teams Adaptive Cards, Outlook calendar holds, and Microsoft Planner entries.

## 10 UNPRECEDENTED INNOVATION POINTS

- **1. Multimodal Context Fusion (OpenCV + Text):** Eliminates context dropouts. When a speaker points to a chart and says, *"Let's fix this trend by Friday,"* standard text tools fail. Botzilla matches that text to the OpenCV-extracted slide snapshot at that exact timestamp, identifying the explicit task visually.
- **2. AI4Bharat Code-Mixed Core:** Built natively for modern workspace realities. It handles mixed-language business logic (Hinglish) with high precision, mapping colloquial statements (e.g., *"Yeh deployment Friday tak complete ho jani chahiye"*) into structured English action items.
- **3. Zero-Trust Localized Privacy Sandbox:** Directly hits the primary evaluation criteria. Before sending data to cloud LLMs, a local regex and spaCy middleware scrubs and anonymizes PII (names, specific budgets, proprietary codes). The real identities are re-injected on the local backend right before UI rendering.

- **4. Forensic Traceability (The "Receipts" Engine):** Eliminates hallucinations completely. Every action item displayed on the dashboard features an interactive "Evidence" tag. Clicking it highlights the corresponding transcript sentence and snaps the video player to the exact second it occurred.
- **5. Staged Agentic Execution Guardrails:** Real-world operations require a human in the loop. The agent acts autonomously by staging Microsoft Graph API configurations as "Pending Actions". The user can visually review, re-assign, or mass-approve the tasks with a single click before execution.
- **6. Pixel-Delta Compute Optimization:** Instead of exhausting cloud compute or running high-latency video models continuously, OpenCV runs light, local math equations on pixel changes, rendering video analysis fast and scalable for standard servers.
- **7. Meeting Health KPI Tracker:** Computes structural engagement analytics. It analyzes talk-time balance per speaker, agenda adherence, and decision density per minute, providing teams with a measurable execution score.
- **8. Cross-Meeting Dependency Mapping:** Tracks carry-over issues. Semantic Kernel logs past tasks to a local vector store. If a blocker from last week's session is brought up again, the agent flags it as a recurring risk item.
- **9. Localized Neural TTS Accessibility Digests:** For impaired or off-site team members, the agent converts the final markdown summaries into short, highly accurate, natural-sounding audio summaries via Neural TTS.
- **10. Deterministic JSON Schema Guardrails:** Prevents arbitrary formatting failures. Every LLM output is strictly bound to a strict validation schema via Pydantic on FastAPI, ensuring the downstream Graph API integrations never drop attributes like dates or owner fields.

## THE "REAL-TIME" ILLUSION: TECHNICAL EXECUTION DETAIL

To provide a premium user experience without the instability of live streams, Botzilla utilizes a highly efficient **Asynchronous Pipeline Processing** strategy.

When a meeting concludes and the user uploads the `.mp4` file, the UI shows a dynamic, real-time progress layout tracking the backend workers:

| Stage             | Processing Engine        | Optimization                                           | Time Required (30m Call) |
|-------------------|--------------------------|--------------------------------------------------------|--------------------------|
| Demuxing          | FFmpeg                   | Direct bitstream copying (no re-encoding)              | ~5 Seconds               |
| Vision Extraction | OpenCV                   | Frame skipping; analyzes only 1 out of every 60 frames | ~15 Seconds              |
| Transcription     | AI4Bharat                | Batch audio slicing passed parallelly to GPU workers   | ~45 Seconds              |
| AI Inference      | Semantic Kernel + GPT-4o | Local caching + precise contextual prompt framing      | ~20 Seconds              |

While the backend processes these items sequentially, the frontend streams the raw transcription text to the user dynamically, making the system feel live. Within roughly 90 seconds, the asynchronous pipeline wraps up, rendering the interactive visual timelines, timestamped slides, and ready-to-fire Microsoft Graph action hooks simultaneously.

Now that the complete structural architecture and unique selling points are defined, MY G, should we focus on writing the specific system prompt rules for Semantic

Kernel to prevent task extraction failures, or should we refine the slide deck outline to showcase these 10 points clearly to the judges?